

Part 3. Tableau, Repository & Open Sharing

Build and publish your Tableau Public dashboard as a publicly accessible open resource.

Prerequisite: Parts 1 & 2 complete | **Level:** Beginner–Intermediate | **Time:** ~3–4 hours

Learning Objectives

- Connect your master CSV to Tableau and build interactive visualizations for SDOH variables.
- Apply dashboard design principles to create a multi-view interactive SDOH dashboard.
- Publish your completed dashboard to Tableau Public as an openly accessible resource.
- Document and share your project through GitHub, create a citable research record with Zenodo when appropriate, and explore institutional repository options for open sharing.

What you need before starting:

- Sign up for a free Tableau Public account at public.tableau.com (takes 2 minutes).
- Your master file: `output/tableau/sdoh_[state]_tableau.csv` from Part 2.
- A modern browser: Chrome or Edge is recommended. Firefox and Safari also work.
- No software download required.

Geographic Unit Used In This Part

Part 2 demonstrates the complete data workflow at the county level using Illinois as the example state. Part 3 uses the same county-level master CSV as its input. All map types, filter formulas, and step-by-step instructions in this part use county as the geographic unit. Students who adapted Scripts in Part 2 for census tract or ZCTA geography can apply the same Tableau steps but will need to adjust the geographic role assignment and FIPS filter formula accordingly.

1. Connect Your Data to Tableau



Corresponding Videos: [Access Tableau](#) & [Connect Your Data](#)

Access Your Free Tableau Public Account

1. Create or sign in to your Tableau Public account
 - First-time user: Go to public.tableau.com and click **Sign Up** in the top right corner. Enter your name, email, and password. Then check your email and click the confirmation link to activate your account.
 - Returning user: Go to public.tableau.com and click **Sign In** in the top right corner. Enter your email and password.
2. After signing in, click your name in the top-right corner, then select **My Profile**. All your published dashboards will appear here.
3. On your profile page, click the blue **Create a Viz** button. This opens the Tableau Public web interface in your browser.

Upload Your CSV and Connect Data

1. The Connect to Data panel opens automatically when you click Create a Viz. Under Files, click Upload from Computer.

2. Navigate to `output/tableau/sdoh_[state]_tableau.csv`, then click Open. Tableau uploads and previews the file.
3. Tableau will display a preview grid of your data rows. Scroll horizontally to verify that all structural tracking columns are present (`fips`, `county`, `svi_overall`, etc.).
4. Locate the FIPS column in the preview. If a numeric symbol (#) appears above the column name, click it and select **Change Data Type to String (Abc)**. FIPS must be text for mapping to work correctly. If you skip this step, the map will not be rendered.
5. Once configured, click **Update Now** and click the Worksheet 1 tab at the bottom to enter the visualization canvas.

Google Drive Alternative

If your CSV is stored in Google Drive, click Google Drive in the Connect panel and sign in. Navigate to your CSV and select it. To update your dashboard with new data later, click Data, then Refresh Data Source in Tableau and republish.

Latitude and Longitude

When you assign the FIPS geographic role, Tableau automatically generates `Latitude (generated)` and `Longitude (generated)` from your FIPS column. These fields mark the county's centroid, which works for filled maps or bubble charts but not for detailed address mapping. For street-level detail, use `tidygeocoder` in R to create precise coordinate columns.

2. Choose Your Map Type

Four Visualization Categories for Regional Data Variance Mapping

When translating SDOH into geographic maps, your choices are driven by data structure and communication goals. The Spatial Visualization Product Matrix below breaks down four common visualization products along two dimensions: Map Typology Focus and System Complexity.

Map Typology Focus refers to what the map emphasizes: specific locations, spatial patterns, or both.

- Reference Map: Where things are (locations, addresses, specific geographic centers). Example: Where are the regional resource centers located?
- Thematic Map: How values vary across space (patterns, rates, statistical differences). Example: Which geographic units have the highest diabetes prevalence?

System Complexity refers to how much data integration, interactivity, and coordination are built into the final product.

- Low Complexity: One dataset with simple interactions. A user can click an isolated point or boundary to see its name, but there is no active filtering. Only one variable is shown at a time.
- High Complexity: Multiple datasets across coordinated views. Filtering one chart automatically updates all other sheets simultaneously using variable selectors and linked highlighting.

(Source: Healthy Regions & Policies Lab)

Spatial Visualization Product Matrix

	Asset map	Thematic map	Dashboard	Story map
Map Type Focus	Reference Mapping	Thematic Mapping	Thematic, Multivariate	Reference and Thematic (narrative sequence)
System Complexity	Low	Low to Medium	High Complexity	Medium to High

Tableau Component	Point/Shape Marks	Filled map (polygon) Bubble map (circle)	Coordinated Multi-Sheet Canvas	Built-in Tableau Story Tab
Data Structure	CSV with Latitude and Longitude	CSV merged to FIPS	Master CSV or relational data source with multiple variables	Master CSV, arranged into narrative sequence
Primary Analytical Goal	Documenting health infrastructure inventory and allocation	Showing how an SDOH variable varies across regions	Exploring multiple SDOH variables simultaneously	Walking an audience through a guided narrative
Toolkit Alignment	HRSA HPSA facility or location layer	County choropleth	The 3-view Tableau dashboard	Final presentation layer linking findings

Dashboard and Story Map are containers as they do not have their own map type. You build Asset Maps and Thematic Maps first (Sections 3 and 4), then assemble them into a Dashboard (Section 5) or Story Map (Section 6).

Map Types Buildable from Your Master CSV

All six map types below can be built directly from `sdoh_[state]_tableau.csv`. Tableau generates latitude and longitude automatically from your FIPS column.

Map Type	Asset Map	Thematic Map	Dashboard / Story	Variables Needed from Master CSV
Simple Map	✓		✓	<code>fips</code> , <code>county_name</code>
Point Distribution Map	✓		✓	<code>fips</code> (Tableau generates lat/lon automatically)
Filled (Choropleth) Map		✓	✓	<code>fips</code> , & any of your numeric variables
Proportional Symbol Map		✓	✓	<code>fips</code> , & Count Variable (e.g., <code>total_enrollment</code>)
Dual-Axis (Layered) Map		✓	✓	Two layers from the same dataset
Filled Map with Pie Charts		✓	✓	<code>fips</code> , & <code>hpsa_prim_care</code> , <code>hpsa_mental_health</code> (reshaped in R)

Map Type Decision Guide

If your research question is...	Use this map type	Category
Which counties have the highest diabetes/poverty/SVI?	Filled (Choropleth) Map	Thematic Map
Where are health care shortage areas located?	Point Distribution Map	Asset Map
How severe are shortages per county?	Proportional Symbol Map	Thematic Map
Which counties have shortages and high diabetes?	Dual-Axis (Layered) Map	Thematic Map
What proportion of shortages are mental health?	Filled Map with Pie Charts	Thematic Map
What is my study area? (reference only)	Simple Map	Asset Map
I want to show multiple SDOH views together	Dashboard	Dashboard
I want to walk through the findings step by step	Story Map	Story Map

Data Structures (polygon vs. point) and Build Steps

The most fundamental choice in Tableau mapping is whether your data uses area boundaries (polygon data) or point locations. Polygon data uses a geographic identifier like FIPS code. Tableau draws the county boundary automatically. Point data requires Latitude and Longitude columns in decimal degrees. You cannot make a choropleth from point data. Tableau can use recognized geographic identifiers such as county FIPS codes to generate centroid points. Facility- or address-level point mapping requires explicit latitude and longitude.

Maps using area boundaries (polygon data) • Simple map — reference overview of locations • Filled (choropleth) map — ratio or aggregated data by area • Filled map with pie charts — composition within each area	Maps using point data (lat/lon coordinates) • Point distribution map — locations and visual clusters • Proportional symbol map — quantitative values at locations • Heatmap (density map) — concentration and trends • Flow (path) map — movement over time • Origin–destination (spider) map — connections between places • Dual-axis (layered) map — two map types combined
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Note: Heatmap, Flow map, and Origin-Destination map cannot be generated from the master CSV in this toolkit. They require individual point data, time-series data, or origin-destination pairs respectively.

3. Asset Maps



Corresponding Video: [All Maps \(Asset Maps\)](#)

Asset maps show where community resources or geographic units are located. They use reference mapping. No data values are encoded in color. Use asset maps to orient your audience to the study area or show where specific resources are located.

Simple Map

A reference map that displays all counties without encoding any data variable in color. Shows where things are. Use it to orient a non-specialist audience to your study area before showing data.

Data Structure Needed

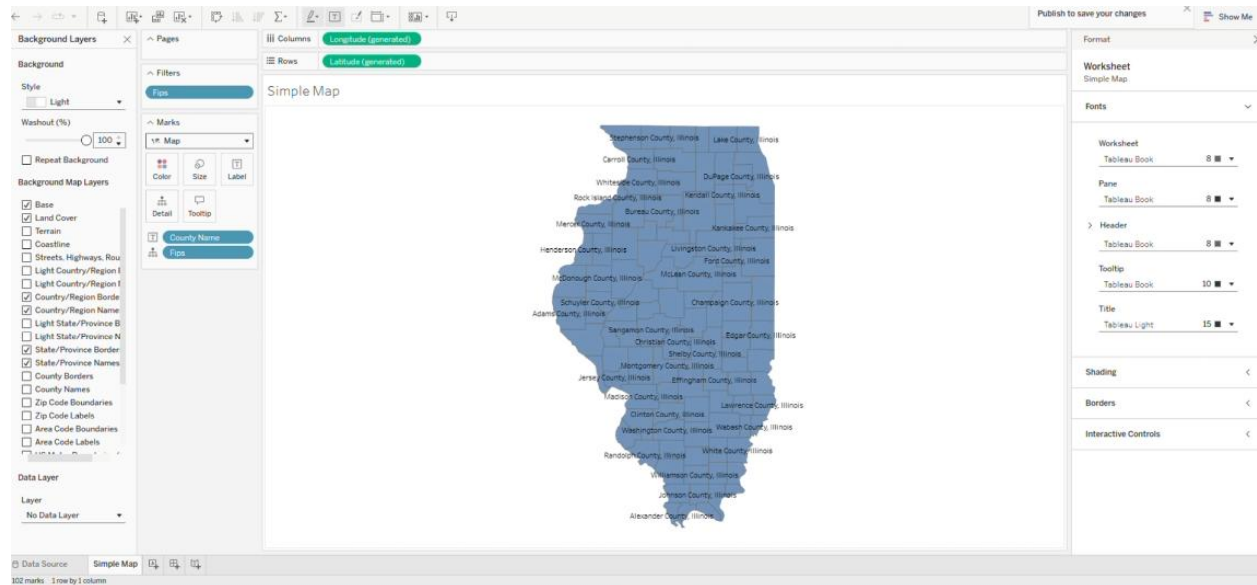
Your master CSV already has this. The column, `fips`, is required:

```
fips    county_name
17001   Adams County, Illinois
17003   Alexander County, Illinois
17005   Bond County, Illinois
...
Required: fips (String, 5-digit)
Optional: county_name (for labels)
```

Step-By-Step: Simple Map

1. Right-click the Worksheet 1 tab and rename it: Simple Map.
2. Right-click the FIPS field in the Data pane, then **Geographic Role**, then **County**. A globe icon shows. Tableau automatically creates Latitude (generated) and Longitude (generated).
3. Double-click the FIPS field. Tableau generates an Illinois map with centroid dots in each county.
4. Drag Fips to the Filters. The filter window will automatically pop up. Click **Condition**, select **By formula**, and enter: `LEFT([Fips], 2) = '17'` for Illinois. Click OK.

5. Under the Marks card dropdown menu, change the Mark type from *Automatic* to **Map**. The map should show IL counties in blue.
6. Drag **County Name** to the **Label** shelf.
7. Go to the top menu bar, click **Format** → **Worksheet** to change the font size to 8pt.
8. Go to the top menu bar, click **Map** → **Background Layers**. In the left pane, set **Washout** to 100%.



Point Distribution Map

A point distribution map plots the locations of entities as individual dots, allowing you to spot visual clusters or regional geographic patterns of occurrence. In public health research, this technique is used to map regional health care shortages, show the distribution of health centers or food pantries, identify spatial clustering of resource deserts in rural areas, and plot educational access points for regional analysis. This toolkit supports two data sources for point distribution maps. Try both to compare what each reveals about your study area.

Data Source 1: County Centroids (Master CSV)

Individual dots indicate the locations of each geographic unit's centroid. Tableau automatically generates coordinates from your FIPS column. Use this map to show where counties with specific characteristics are located.

Data Structure Needed

The column **Fips** is required with any variable (e.g., poverty rate) for color or size.

name	latitude (generated)	longitude (generated)	poverty rate
Adams County	39.9217	-91.1849	14.5
Alexander Co.	37.1481	-89.3201	22.0
Bond County	38.8922	-89.4323	8.0
...	...	...	...
Required: Latitude (decimal, Measure)			
Required: Longitude (decimal, Measure)			
Optional: any variable for Color or Size			

Step-By-Step: County Centroids

1. Click the Sheet tab and rename it: Point Distribution Map.
2. Double-click the **FIPS** field. Tableau will automatically generate a U.S. map with one centroid dot per county.
3. Drag FIPS to the Filters shelf. Click the **Condition** tab, select By formula, and enter: `LEFT([Fips],2) = '17'`. Click OK.
4. In the Marks card dropdown menu, change the mark type from *Automatic* to **Circle**.
5. Drag **County Name** from the Data pane and drop it onto the **Detail** tile on the Marks card. This makes the field available for your tooltip.
6. Drag **poverty_rate** (or any SDOH variable) to the **Color** shelf. Click **Color**, then set Opacity to 70%.
7. Click the **Tooltip**. Delete the default text in the editor box. Use the **Insert** dropdown menu in the top-right corner of the window to add your fields, structuring the text like this: `County: <County Name> | Poverty: <SUM(Poverty_Rate)>%`

Data Source 2: Actual Health Center Locations (HRSA File)

Individual dots mark HRSA-funded health center sites with precise street-level coordinates. Each dot represents one health center. Best for showing where specific facilities are located within and across counties.

Download Before Building:

Go to data.hrsa.gov/data/download and scroll to **Health Center Service Delivery and Look-Alike Sites**. Download the CSV (updated daily, free, no login required). Save as `data/raw/hrsa_health_centers.csv`. This file already contains latitude and longitude, thus upload it directly to Tableau as a second data source. No R processing required.

Data Structure Needed

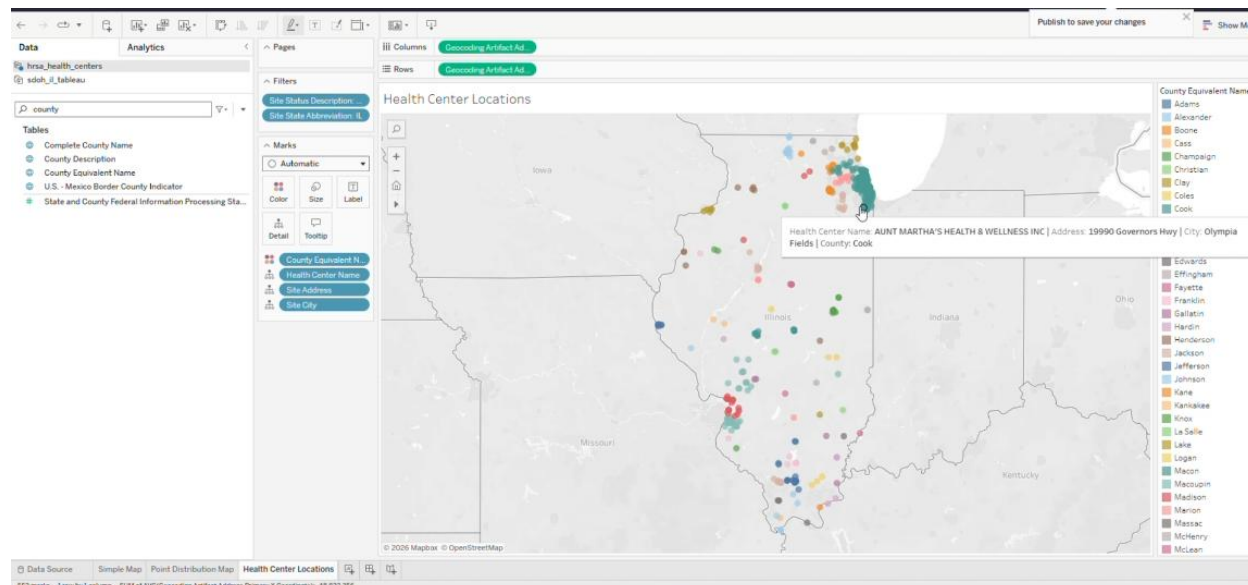
Latitude (Measure) and longitude (Measure) are required.

site name	city	county	state abbreviation	latitude	longitude
Heartland Health	Peoria	Peoria County	IL	40.6936	-89.5890
ACCESS Community	Chicago	Cook County	IL	41.8781	-87.6298

Step-by-Step: Actual Health Center Locations

1. In Tableau, click the **Data** menu, then **New Data Source**, then **Upload from Computer**. Upload `hrsa_health_centers.csv`. Tableau loads the full national file.
2. In the Data pane, right-click **Geocoding Artifact Address Primary X Coordinate**, hover over **Geographic Role**, and select **Longitude**. Then, right-click **Geocoding Artifact Address Primary Y Coordinate**, hover over **Geographic Role**, and select **Latitude**. (This tells Tableau to load the real U.S. base map).
3. Click **Update Now** to complete the data load.
4. Click the Worksheet tab + to open a new sheet. Rename it: Health Center Locations.
5. In the top-left of the Data pane, make sure `hrsa_health_centers.csv` is selected.
6. Drag **Geocoding Artifact Address Primary X Coordinate** to **Columns**. Drag **Geocoding Artifact Address Primary Y Coordinate** to **Rows**. Tableau switches to a map view with one dot per health center site across the entire United States.
7. Go to the top menu bar, click **Analysis**, and uncheck **Aggregate Measures**. This forces Tableau to display every individual health center point instead of adding them together on a blank grid.
8. Drag **Site Status Description** to the Filters shelf. Select **Active**. Click OK.
9. Drag **Site State Abbreviation** to the Filters shelf. Select **IL (or your state abbreviation)**. Click OK. The map zooms to show only your state's active health centers.

10. In the Marks card dropdown menu, change the mark type from *Automatic* to **Circle**.
11. Drag Health Center Name, Site Address, and Site City from the Data pane onto the **Detail** tile of the Marks card so they are accessible in your tooltip text.
12. Drag County Equivalent Name to the Color shelf to distinguish sites by county. Click the Color card and set the Opacity slider to 70%.
13. Click the **Tooltip** card. Delete the default text in the editor window. Use the **Insert** dropdown in the top-right corner to add your loaded fields, formatting the line to read: Site: <Health Center Name> | Address: <Site Address> | City: <Site City> | County: <County Equivalent Name>



If you have a CSV with street addresses for facilities like food pantries, community clinics, or social service sites, you can convert those addresses to latitude and longitude coordinates using the `tidygeocoder` package in R. Once geocoded, upload the resulting CSV to Tableau as a second data source and follow the same build steps as Data Source (HRSA File) above.

```
# install.packages("tidygeocoder")
library(tidygeocoder)
library(tidyverse)

# Load your facility file with a street address column
my_facilities <- read_csv("data/raw/my_facilities.csv")

# Geocode using free OpenStreetMap service - no API key needed
# Change 'street address' to match your actual column name
my_facilities_geo <- my_facilities %>%
  mutate(full_address = paste(street_address, city, state, "USA")) %>%
  geocode(full_address,
          method = "osm",
          lat = latitude,
          long = longitude)

cat("Geocoded:", nrow(my_facilities_geo), "rows\n")
cat("Missing coordinates:", sum(is.na(my_facilities_geo$latitude)), "\n")

write_csv(my_facilities_geo, "output/tableau/my_facilities_geocoded.csv")
```

4. Thematic Maps



Corresponding Video: [All Maps \(Thematic Maps\)](#)

Thematic maps show how data values vary across geographic areas. They are the most common type of map in SDOH research. Each map type encodes data differently. You need to choose based on your research question and data structure.

Filled (Choropleth) Map

A choropleth map shades areas by data value, with darker tints indicating higher or lower concentrations. It is most common in public health research. This visualization shows how variables such as diabetes prevalence, poverty rates, or SVI values vary across counties simultaneously, helping identify geographic clusters of high or low health burden, compare county metrics to the state average, and map any continuous percentage, index, or ranked variable.

Data Structure Needed

Your master CSV. Two minimum columns, including `fips` and any numeric variables

fips	diabetes_pct	poverty_rate	svi_overall
17001	11.2	14.3	0.62
17003	17.8	29.7	0.91
17005	10.4	11.2	0.44
...	...	...	...

Required: `fips` (String, 5-digit)

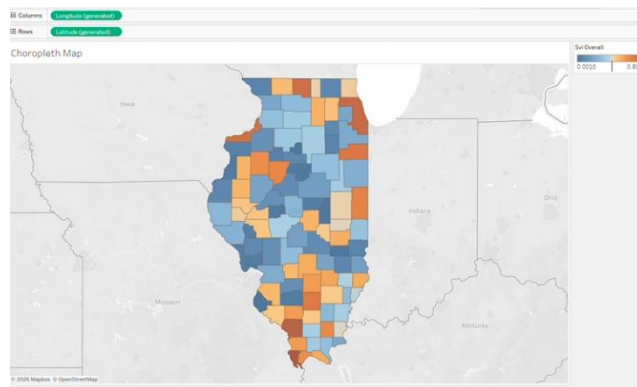
Required: any numeric measure variable

Choosing The Right Color Palette

- Sequential (single-color ramp): Use for variables that move from lower to higher values, such as poverty rates, SVI percentiles, or EJI percentiles. Example: `poverty_rate` (white to dark blue). In Tableau: Edit Colors, then choose a sequential palette such as Blue Sequential.
- Diverging (two-color ramp): Use when a variable has a meaningful midpoint or reference value and you want to show values above and below that point. In Tableau: Edit Colors, then choose an appropriate diverging palette.
- Never use rainbow: Rainbow color scales are not accessible for many colorblind users and do not convey order clearly. Use a sequential or diverging palette appropriate for the variable.

Step-by-step: Filled (Choropleth) Map

1. Double-click the sheet tab and rename it: Choropleth Map.
2. Right-click FIPS in the Data pane, hover over **Geographic Role**, and select **County**. A globe icon will appear next to the field.
3. Double-click the FIPS field. Tableau will generate a U.S. map displaying individual centroid dots.
4. Drag FIPS to the Filters shelf. In the pop-up prompt, select All values and click Next. Click the **Condition** tab, select **By formula**, and enter: `LEFT([Fips], 2) = '17'` for Illinois. Click OK to zoom into your state.
5. Change the mark type dropdown on the **Marks card** from *Automatic* to **Map**. The dots will instantly transform into solid, filled county shapes.
6. Drag your variable (e.g., `svi_overall`) from the Data pane and drop it directly onto the **Color** shelf on the **Marks card**. The counties will immediately shade by value.
7. Click the **Color** card and select **Edit Colors**. Choose Blue Sequential for burden/risk variables, or Orange-Blue Diverging for index variables (like SVI or EJI). Click OK. (Note: The web interface automatically syncs to your data's true minimum and maximum values).
8. Drag `County Name` and your variable from the Data pane and drop them onto the **Detail** tile of the Marks card. This loads them into background memory so your tooltips can read them.
9. Click the **Tooltip** card. Delete the default text in the editor window. Use the **Insert** dropdown menu in the top-right corner to add your loaded fields, formatting the text to read:
`County: <County Name> | Variable Name: <SUM(Variable)>`
10. Verify that all counties are shaded, no areas show Null or gray, the color legend shows the correct range, and the tooltip works on hover.



If your map shows dots instead of shaded counties, click the mark type dropdown in the Marks card and change it to Map. If the Map is unavailable, return to the Data Source tab, confirm that FIPS is a string, and reassign the County geographic role.

Proportional Symbol Map

A proportional symbol map uses circles scaled to data values at each location to show both position and magnitude simultaneously. It reveals where issues exist and their significance, ideal when absolute size matters over rate. In public health, it's used for counts of shortages, uninsured individuals, or food pantry visits and social service referrals.

Data Structure Needed

From your master CSV file, three variables are needed (1) `fips` (Tableau generates lat/lon automatically), (2) count or total variable for the Size shelf (e.g., `total_enrollment`), and (3) rate variable for the Color shelf (e.g., `poverty_rate`).

name	latitude(generated)	longitude(generated)	total_enrollment	poverty_rate
Adams County	39.9217	-91.1849	330	14.5
Alexander Co.	37.1481	-89.3201	502	22.0
Bond County	38.8922	-89.4323	135	8.0

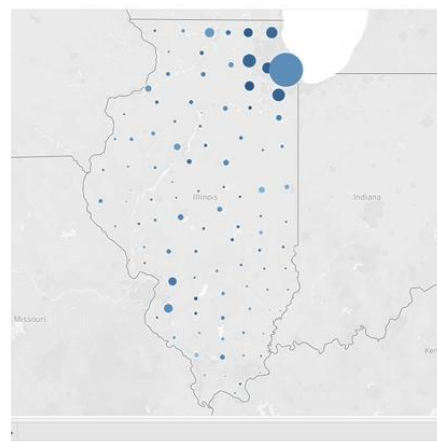
Required: Latitude, Longitude (Measures)
 Required: count/total variable for Size shelf

Optional: rate or score variable for Color shelf
 Key distinction: Size = count, Color = rate

Step-by-step: Proportional Symbol Map

1. Double-click the sheet tab and rename it: Proportional Symbol Map.
2. Right-click FIPS in the Data pane, hover over **Geographic Role**, and select **County**. A globe icon will appear next to the field.
3. Double-click the FIPS field. Tableau will automatically generate a U.S. map layout with individual centroid dots.
4. Drag FIPS to the **Filters** shelf. In the pop-up window, click the **Condition** tab, select **By formula**, and enter: `LEFT([Fips], 2) = '17'` for Illinois. Click OK to zoom into your state.
5. Change the mark type dropdown on the **Marks card** from *Automatic* to **Circle**.
6. Drag your count variable (e.g., `total_enrollment`) from the Data pane and drop it onto the **Size** shelf on the Marks card. The circles will immediately scale proportionally to the data values.
7. Drag your rate variable (e.g., `poverty_rate`) from the Data pane and drop it onto the **Color** shelf on the **Marks card**. Your map now displays magnitude by size and intensity by color.
8. Drag `County Name` from the Data pane and drop it onto the Detail tile of the Marks card so it is available in your tooltip memory.
9. Click the **Size card**. Adjust the slider slightly to the right to make the circles larger, ensuring the smallest circles remain clearly visible and the largest circles do not completely obscure the map background.
10. Right-click the title of your size legend on the right side of the screen and select **Edit Title**. Change it to match your count field (e.g., Total Enrollment). Do the same for your color legend (e.g., Poverty Rate (%)).
11. Click the **Tooltip card**. Delete the default text in the editor window. Use the Insert dropdown menu in the top-right corner to add your loaded fields, formatting the text to read cleanly:

```
County: <County Name> | Enrollment:
<SUM(total_enrollment)> | Poverty:
<SUM(poverty_rate)>%
```
12. Verify that your state's counties display circles of varying sizes, the color ramp accurately reflects the data range, and the tooltip displays the correct county information upon hover.



Dual-Axis (Layered) Map

A dual-axis layered map overlays two map types on one canvas. This combo allows for simultaneous viewing of regional patterns and local magnitudes. It's helpful for assessing health outcomes in relation to problem sizes, for example, by shading counties based on diabetes rates and adding symbols for health professional shortages over a social vulnerability index.

Data Structure Needed

Two layers are combined in Tableau: Layer 1 (choropleth) uses `fips` and any rate variable (such as `diabetes_pct`), while Layer 2 (symbols) uses `fips` and a count variable.

```

Layer 1 (choropleth base):
fips      diabetes_pct  svi_overall
17001      11.2          0.62
Source: sdoh_il_tableau.csv

Layer 2 (symbols):
name      latitude(generated)  longitude(generated)  total_enrollment
Adams County  39.9217          -91.1849              350
Source: sdoh_il_tableau.csv

```

Step-by-step: Dual-Axis (Layered) Map

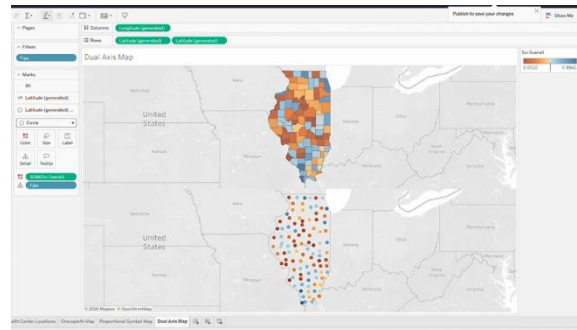
To build this combination, follow these exact steps:

Part 1: Build the Background Layer

1. Click your sheet tab at the bottom and rename it: Dual Axis Map.
2. Right-click FIPS in the Data pane, hover over **Geographic Role**, and select **County**
3. Double-click the FIPS field to generate the base map layout.
4. Drag FIPS from the Data pane and drop it onto the **Filters** shelf. Click the **Condition** tab, select **By formula**, and enter: `LEFT([Fips], 2) = '17'` for Illinois. Click OK.
5. Change the mark type dropdown on the **Marks card** from *Automatic* to **Map** to create your solid filled counties.
6. Drag `svi_overall` from the Data pane and drop it directly onto the **Color** shelf.
7. Click the **Color card**, select **Edit Colors**, and choose a diverging palette like **Orange-Blue Diverging** (since SVI is an index variable). Click OK.

Part 2: Layer the Circles on Top

8. Hold the **Ctrl** key on your keyboard, click the green **Latitude (generated)** pill on the Rows shelf, and drag it to the right to duplicate it. Your screen will split into a top and bottom map.
9. Click the **Latitude (generated) (2)** section on your **Marks card** to open the controls for your second layer.
10. Change the mark type dropdown for this second layer from *Map* to **Circle**.
11. Drag `svi_overall` off this second Marks card to clear it out.
12. Drag `diabetes_pct` from the Data pane and drop it onto the **Size** shelf. The circles will now scale based on diabetes prevalence.
13. Drag `County Name` from the Data pane and drop it onto the **Detail** tile to load it into tooltip memory.
14. Go back to the **Rows** shelf at the top, right-click the second green **Latitude (generated)** pill, and select **Dual Axis** to merge the dots onto your filled background.
15. Click the **Color** tile on your circle **Marks card** and set the **Opacity to 70%** so you can see the SVI colors underneath your diabetes circles.



Filled Map with Pie Charts

A map overlays pie charts on county centroids over a choropleth map to show categorical breakdowns across regions. It reveals both overall rates and internal composition at once but should be used for only two to three categories, as smaller pie charts become hard to read. In public health, it helps visualize

resource or burden distribution within counties, like health shortage types or other categorical metrics across counties.

Data preparation in R (required before building in Tableau)

Your master CSV has columns for `hpsa_prim_care`, `hpsa_dental`, and `hpsa_mental_health`. Reshape it to long format before importing into Tableau.

```
# Data preparation in R (required before building in Tableau)
# Run this in R after completing Part 2 Scripts #1-#6

library(tidyverse)

# ——— Load master CSV ———
STATE_LABEL <- "il" # change to match your state
master <- read_csv(paste0("data/master/sdoh_", STATE_LABEL, "_master.csv"),
  col_types = cols(fips = col_character()))

# ——— Reshape HPSA columns to long format for Tableau Pie Chart Map ———
# Note: "count" stores the HPSA designation level:
# 0 = none, 1 = partial, 2 = full
hpsa_long <- master %>%
  select(fips, county_name, hpsa_prim_care, hpsa_dental, hpsa_mental_health) %>%
  pivot_longer(
    cols = c(hpsa_prim_care, hpsa_dental, hpsa_mental_health),
    names_to = "shortage_type",
    values_to = "count"
  ) %>%
  mutate(shortage_type = case_when(
    shortage_type == "hpsa_prim_care" ~ "Primary Care",
    shortage_type == "hpsa_dental" ~ "Dental",
    shortage_type == "hpsa_mental_health" ~ "Mental Health"
  ))

cat("Rows in reshaped file:", nrow(hpsa_long), "\n")
cat("Expected:", nrow(master) * 3, "(3 shortage types x", nrow(master),
  "counties)\n")

write_csv(hpsa_long, "output/tableau/hpsa_long.csv")
cat("Saved: output/tableau/hpsa_long.csv\n")
```

Note. count represents the HPSA designation level

Data Structure Needed

Long-format data with one row per county per category:

fips	shortage_type	count
17001	Primary Care	2
17001	Dental	1
17001	Mental Health	1
17003	Primary Care	3
17003	Mental Health	2
...	...	...

Required: fips (String), category column, count
 Format: long (one row per county-category pair)
 Create in R: `hpsa_clean %>% pivot_longer(...)`

Step-by-step: Filled Map with Pie Charts Using Two Datasets

Part 1: Connect the Two Datasets

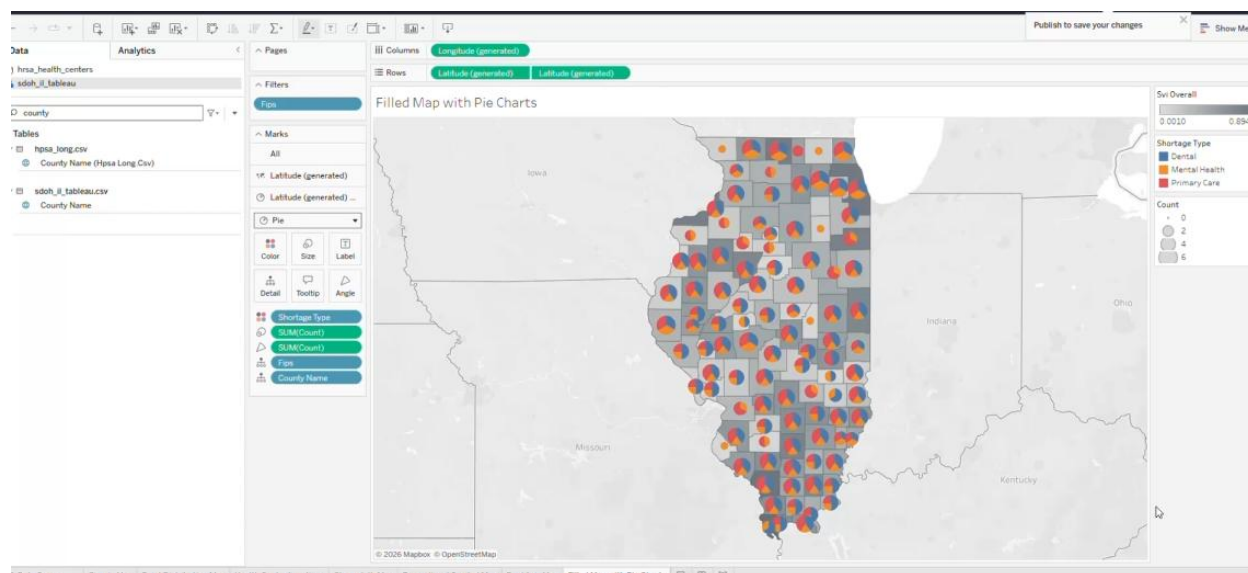
1. Click the **Data** menu, select **New Data Source**, click **Upload from Computer**, and upload `sdoh_[state]_tableau.csv`.
2. Click the small + (**Add**) icon next to the **Connections** header. Select **Upload from Computer** and upload `hpsa_long.csv`.
3. Drag the `hpsa_long.csv` file from the left panel and drop it to the right of the `sdoh_[state]_tableau.csv` box.
4. Set the matching fields in the relationship mapping box that pops up at the bottom:
 - Under `sdoh_[state]_tableau`, select **Fips**.
 - Under `hpsa_long.csv`, select **Fips**.
5. Click the **Sheet** tab in the bottom-left corner to open your workspace canvas.

Part 2: Build the Background Map Layer

6. Click the arrow next to `sdoh_[STATE]_tableau.csv` to expand it. Ensure that all variables for this background layer are sourced from your master CSV file.
7. Click **Fips** in the Data pane, change data type to **String**, hover over **Geographic Role**, and select **County**.
8. Double-click the **Fips** field to automatically generate your baseline map layout using Tableau's built-in coordinates.
9. Drag **Fips** from the Data pane onto the **Filters** shelf. Click the **Condition** tab, select **By formula**, and enter: `LEFT(STR([Fips]), 2) = '17'` to cleanly isolate Illinois. Click **OK**.
10. Change the mark type dropdown menu on the Marks card from *Automatic* to **Map** to create solid filled county shapes.
11. Drag `Svi Overall` from the Data pane and drop it onto the **Color** shelf of the Marks card. Click **Color** → **Edit Colors**, select the **Gray** palette

Part 3: Layer the Pie Charts Using a Dual Axis

12. Hold the **Ctrl** key on your keyboard, click the green **Latitude (generated)** pill sitting on your **Rows** shelf, and drag it to the right to duplicate it. Your screen will split into a top and bottom map.
13. Click the lower section of your split Marks card labeled **Latitude (generated) (2)** to isolate the settings for your second layer.
14. Drag the green `Svi Overall` pill off this second Marks card to clear background shading from your charts.
15. Change the mark type dropdown on this second Marks card from *Map* to **Pie**.
16. Expand `hpsa_long.csv`. All variables for this layer come from this dataset.
17. Drag **Shortage Type** from the Data pane and drop it onto the **Color** tile of this Marks card to define your individual pie slices.
18. Drag **Count** onto the **Angle** to control slice size. Drag **Count** onto the **Size** to scale the pie diameter by the shortage volume.
19. Go back up to your top **Rows** shelf, right-click the second green **Latitude (generated)** pill, and click **Dual Axis** to snap your pie charts directly over your shaded county background map.
20. Click the **Size** on the pie Marks card and adjust the slider until the charts are visible without overlapping borders. Drag `County Name` to Detail.
21. Click the **Tooltip** card on the pie Marks card, delete the default text, and use the **Insert** dropdown menu to format your clean label:
County: <County Name> | Shortage Type: <Shortage Type> | Designation Count: <Count>



5. Assemble Your Dashboard



Corresponding Video: [All Maps \(Thematic Maps\)](#)

A Dashboard combines multiple coordinated views, any combination of the maps you built in the previous sections, into one interactive canvas. Users can filter one view, and all others update automatically.

Choose Which Views to Include

Your dashboard must include at least three views. Choose from any maps you built in the previous sections. The recommended combination for SDOH research is:

- View 1 (primary): Filled (Choropleth) Map of your outcome variable, showing WHERE the burden is highest.
- View 2 (supporting): Ranked Bar Chart of your outcome variable, showing WHICH counties rank worst.
- View 3 (supporting): Scatter Plot of predictor vs outcome, showing the SDOH relationship.

You can substitute or add any map. For example, replace the bar chart with a Point Distribution Map showing HPSA shortage locations or add a Proportional Symbol Map as a fourth view.

Build Your Ranked Bar Chart

The ranked bar chart shows every county sorted from highest to lowest by your outcome variable.

1. Upload `sdoh_[state]_tableau.csv`. Click your sheet tab at the bottom and rename it: Bar Chart.
2. Drag `County Name` from the Data pane to the Columns shelf, then drag `[OUTCOME_Variable]` from the Data pane to the Rows shelf to create your vertical bars.
3. Sort the chart in descending order by clicking the **Sort Descending** icon in the top toolbar (the button showing vertical bars with a down arrow).
4. Click the **Analytics** tab next to the Data pane. Drag **Average Line** onto the canvas and drop it directly onto the **Table** box that appears. Right-click the newly created line, select **Edit**, and change the **Label** dropdown to **Custom**, typing: `State Average: <Value>`.

5. Drag [PREDICTOR_Variable] to the **Color** on the Marks card. Darker bars will now represent higher predictor values.

Build Your Scatter Plot

The scatter plot shows the relationship between your predictor variable on the X axis and your outcome variable on the Y axis where each dot represents an individual county.

1. Click your sheet tab at the bottom and rename it: Scatter Plot.
2. Drag [PREDICTOR_Variable] to the **Columns** shelf and [OUTCOME_Variable] to the **Rows** shelf.
3. Click **Analysis** in the top menu, then **uncheck Aggregate Measures**.
4. Click the **Analytics** tab, drag **Trend Line** to canvas, then drop it into the **Linear** box.
5. Right-click your **X-axis scale**, select **Edit Axis...**, and set the **Axis Title** to your predictor's label. Repeat for the Y-axis.

Dashboard Layout Design

Dashboard layout guides the reader's eye and clarifies view relationships. A poorly designed dashboard feels overwhelming, even with good data. Here are principles used by professional SDOH dashboard designers.

Principle (1) Lead with the map, support with charts

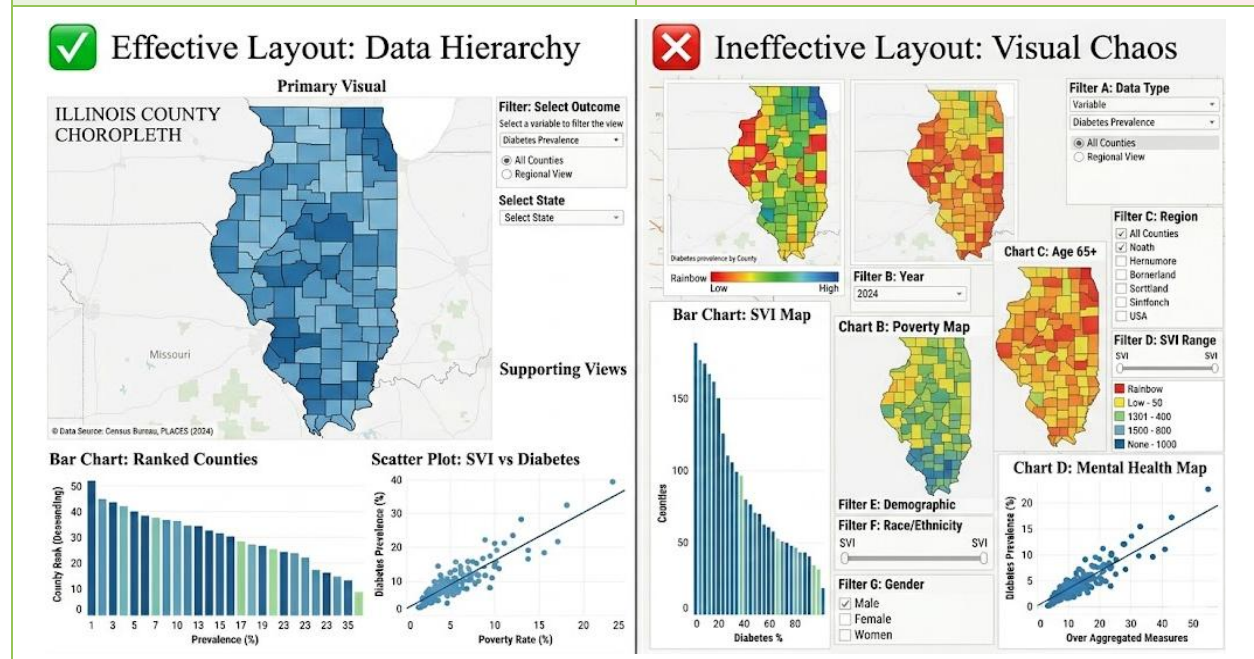
The choropleth map shows “where,” being the most visually compelling element, usually in the largest area on top or left. Supporting charts (bar, scatter) indicate “how much” and “what relationship,” placed smaller below or to the right.

✓ Effective layout

- Large choropleth map in the top or left position
- Bar chart below or to the right
- Scatter plot below or to the right
- Filter panel narrow, on the right edge
- Title and source note at top and bottom

✗ Ineffective layout

- Four charts all the same size competing equally
- Map squeezed into a small corner
- Bar chart taller than the map
- Filters scattered across the canvas
- No title, no data source note



Principle (2) Use filter actions, not separate dashboards

Filter actions let users click one view to update all others simultaneously. Clicking a geographic unit on the map highlights the same unit in the bar chart and scatter plot. Design your dashboard so each view is linked to all others. Instructions for adding filter actions are in the Assemble the Dashboard Canvas section below.

Principle (3) One variable per map, multiple variables across views

A choropleth map shows a single variable using color. Adding a second variable with size or shape causes visual overload. Instead, show one variable per map and use a scatter plot to display the relationship between two variables.

Dashboard contains:

Map: diabetes_pct (color only)

Scatter: poverty_rate (X) vs diabetes_pct (Y)

Bar: counties ranked by diabetes_pct

✓ Each view has one job. They answer: where is diabetes most prevalent, which counties rank highest, and is poverty associated with it?

Dashboard with variables on one map using color, size, and shape:

Map: diabetes_pct (color), poverty (size), and SVI (shape)

✗ Visually overwhelming. The reader cannot accurately perceive three visual channels at once. Each variable competes for attention.

Principle (4) Size communicates priority

In Tableau, the size of each view on the dashboard canvas signals its importance to the reader. Use size deliberately:

- Largest view (50–60% of canvas): Your primary finding is usually shown on the choropleth map, which you want the reader to see first.
- Medium views (20–25% each): Supporting context. Charts and plots illustrating map patterns.
- Narrow strip or small panel: Filter controls, legend, parameter selectors. Keep them small.
- Full-width banners (very narrow): Title text and data source note. It needs to be easy to find when readers look for it.

Principle (5) Title every view, cite every data source

Every sheet on the dashboard needs its own title explaining what it shows. The dashboard itself needs a source note at the bottom. Untitled views force the reader to guess what they are looking at.

✓ Good sheet title format

- Choropleth map: "Diabetes Prevalence by County (%)"
- Bar chart: "Illinois Counties Ranked by Diabetes Prevalence"
- Scatter: "Poverty Rate vs. Diabetes Prevalence, Illinois Counties"
- Source note: "ACS 2020–2024 · CDC PLACES 2025 · SVI 2022"

✗ Poor sheet title format

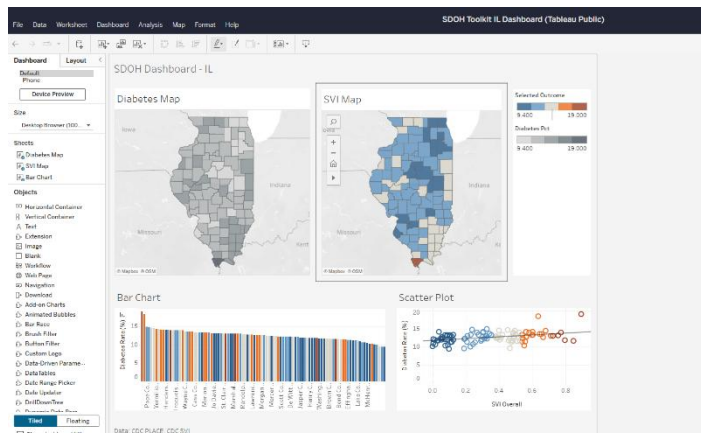
- "Map" or "Sheet 1"
- "Chart" or no title at all
- "Scatter Plot"
- No data source note anywhere on the dashboard



Corresponding Video: [Assemble Your Dashboard](#)

Assemble the Dashboard Canvas

1. Click the **New Dashboard** icon at the bottom of the screen.
2. Drag a **Vertical Container** from the left panel onto the empty canvas.
3. Drop your **map** sheet into the top of the canvas and drag its bottom edge until it occupies 55% to 60% of the dashboard height.
4. Drag a **Horizontal Container** from the left panel and drop it directly underneath the map.
5. Drop your **Bar Chart** into the left side of the horizontal container and your **Scatter Plot** into the right side.
6. Drag a **Text** box to the very top of the dashboard and type your title using the formatting from Principle 5 at 16pt and regular weight.
7. Drag a **Text** box to the very bottom of the dashboard and list your active public health datasets in 10pt gray font.
8. Select **Dashboard** from the top menu, click **Actions**, select **Add Action**, choose **Filter**, and set the run action option to **Select** to link your map, bar chart, and scatter plot.
9. Click a county on the map to confirm that both the bar chart and scatter plot instantly filter to isolate that same area, then click a blank space to reset the view.



Save and Publish to Tableau Public

1. Click the blue **Publish As** button in the top right corner of the screen to open the save dialog window.
2. Type **SDOH-Toolkit-[State]-Dashboard** in the workbook name field and click the blue **Publish** button.
3. Wait for the browser to load your newly published dashboard page on the Tableau Public website.
4. Close the editing window. The window will show the Tableau Public page.
5. Click the **Share** icon in the top-right corner to open distribution settings. Copy the URL from the **Link** box for direct sharing. Copy the HTML code from **Embed Code** to embed the dashboard into a website or report.

6. Build Your Story Map



Corresponding Video: [Story Map](#)

A Tableau Story uses the built-in story feature to arrange sheets and dashboards into a guided narrative. Unlike a Dashboard, which shows all views at once, a Story displays one view with annotations, guiding readers through your research, data, and findings step by step.

Story Map vs. Dashboard

Dashboard	Story Map
Shows multiple views at once	Shows one view at a time in guided sequence
Users explore with filter actions	Author guides users through a specific argument
Best for: data exploration, public reporting	Best for: presentations, policy briefs
Built from any map views in Sections 3 and 4	Built from same sheets

Choose Which Views to Include in Your Story

A story can include any sheet or dashboard you have already built. A recommended five point narrative for socio-spatial health research includes the following structure.

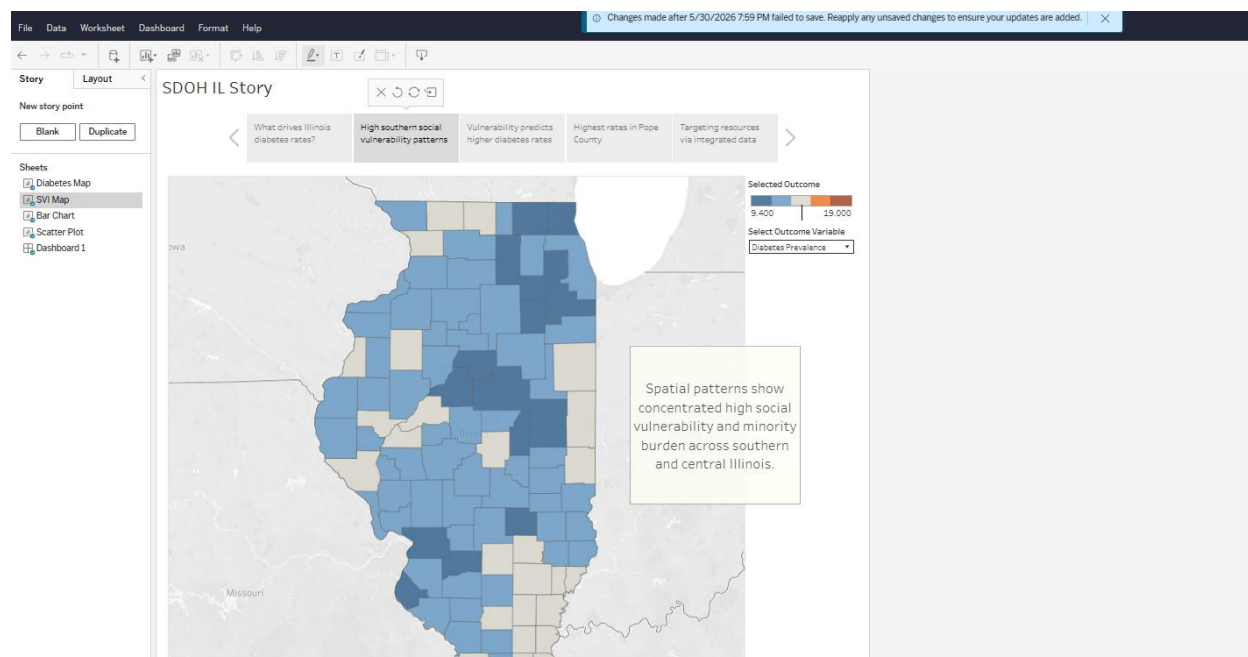
- Story point 1. Research question to show your outcome variable map.
- Story point 2. Strongest predictor map to illustrate social vulnerability.
- Story point 3. Scatter plot with a trend line to show their mathematical relationship.
- Story point 4. A ranked bar chart to demonstrate specific county-level disparities.
- Story point 5. Completed a full dashboard to give a holistic, interactive view.

Step-by-step: Build Your Story

1. Click the **New Story** icon at the bottom of the screen to open the story workspace canvas.
2. Build each of your five story points using the layout matrix below, clicking the **Blank** button in the left panel to generate each new slide.

Story Point	Caption	Map/Figure Sheet	Text Box
Story Point 1. Research question	What drives Illinois diabetes rates?	Drag your outcome Map sheet from the left pane.	
Story point 2. Predictors	High southern social vulnerability patterns	Drag your Predictor Map sheet from the left pane.	Spatial patterns show concentrated high social vulnerability and minority burden across southern and central Illinois.
Story point 3. The relationship	Vulnerability predicts higher diabetes rates	Drag your Scatter Plot sheet from the left pane.	Linear regression shows a positive link: higher social vulnerability predicts higher diabetes rates, supported by the upward trend.
Story point 4. Geographic patterns	Highest rates in Pope County	Drag your Bar Chart sheet from the left pane.	Geographic differences are stark, with southern rural areas like Pope County showing the highest diabetes rates at over 15 percent.
Story point 5. Conclusion	Targeting resources via integrated data	Drag your full SDOH Dashboard-IL from the left pane.	This dashboard enables public health practitioners to cross-reference geographic disease burdens with social vulnerability, optimizing resource allocation and intervention targeting.

3. Click through each narrative tab at the top of the story grid to verify that all five points display their respective caption text, visualizations, and text boxes properly.
4. Click the blue **Publish** button in the top right corner of the screen to save your complete data story online.



7. Adapt and Share



Corresponding Video: [Adapt and Share](#)

Change Variables on an Existing Map

Instead of building a brand-new sheet from scratch, you can quickly swap out the data on your current map or add a dynamic dropdown menu for your users.

Method A: Swap directly (fastest)

- Open your map sheet. In the Marks card, find the green pill on the Color shelf.
- From the Measures list, drag your new variable directly onto the existing green pill. Tableau swaps the variable and re-shades the map immediately.
- Update the color palette, tooltip, and sheet title to match the new variable.

Method B: Add a parameter (lets users switch variables)

1. Click the small drop-down arrow next to the search bar in the Data pane and select **Create Parameter**.
2. Name the parameter `Select Outcome Variable` and change the **Data type** dropdown to **String**. Select **List** under the **Allowable values options** to open the choices grid.
3. Type your exact raw database variable name (e.g., `Diabetes Pct`) in the **Value column** and type your label (e.g., `Diabetes Prevalence`) in the **Display As column**.
4. Type your second raw database variable name (e.g., `Depression Pct`) in the next row under the **Value column** and type your label (e.g., `Depression Prevalence`) in the **Display As column** before clicking OK.
5. Click the drop-down arrow next to the search bar again and select **Create Calculated Field**. Name the calculation, `Selected Outcome`, and type the matching logic formula:

```
IF [Select Outcome Variable] = 'Diabetes Pct' THEN [Diabetes Pct]
ELSEIF [Select Outcome Variable] = 'Depression Pct' THEN [Depression Pct]
```

```
ELSE NULL
END
```

6. Drag your new **Selected Outcome** calculated field from the Data pane and drop it directly on top of the existing color variable on your Marks card to replace it.
7. Hover over “**Select Outcome Variable**” under the Parameter at the bottom of the Data pane. Click the down arrow, select **Show Parameter** to add the interactive dropdown menu to your workspace.

Export Maps for Reports or Presentation/Publication

Method A: Download from Tableau Public • Navigate to your final published dashboard link on the Tableau Public website. • Click the Download icon located in the upper right toolbar area of the published webpage layout. ○ Select Image from the format options menu to save a flat picture file. ○ Select Workbook instead if you want to save the raw twbx pack file to your computer.	Method B: Browser Screenshot • Press the Windows key + Shift + S on a Windows keyboard to open the snipping menu. • Press the Command key + Shift + 4 on a Mac keyboard to activate the crosshair crop utility. • Drag your cursor selection box to isolate just the active visualization frame while completely leaving out the outer web browser address bars.
--	---

Adapt the Dashboard for Another State

Because you enabled downloads, any student can adapt your dashboard.

```
Step 1: Go to the published dashboard. Click the download icon. Select Tableau
Workbook.
Step 2: Go to public.tableau.com. Sign in. Click Create a Viz.
Step 3: Upload from Computer. Upload the .twbx file you downloaded.
Step 4: Confirm FIPS is String (Abc) in the Data Source pane.
Step 5: Update the FIPS filter. Change 'starts with 17' to your state prefix.
        (Indiana = 18, Wisconsin = 55, Missouri = 29, Michigan = 26)
Step 6: Update dashboard titles to reflect your state and variables.
Step 7: File > Save to publish under your own Tableau Public profile.
```

Troubleshooting Common Tableau Problems

These are the most common issues students encounter when building SDOH maps in Tableau, with specific fixes.

Problem	What you see	Fix
Map does not render / shows blank	Nothing appears, or a world map displays without any geographic units.	Go to Data Source tab. Confirm FIPS shows Abc (string variable). Right-click FIPS, then Geographic Role, then County. Return to the sheet and double-click FIPS again.
Only some counties appear (others gray)	Most counties are shaded correctly, but some are displayed in gray.	Gray means missing data for that county. Run filter(master, is.na(your_variable)) in R to identify which geographic units have missing values.
Map shows entire U.S. not just your state	All 3,000+ U.S. counties appear	Edit the FIPS filter. Click the Filter → Edit Filter → Condition , select By formula and type: <code>LEFT([Fips],2) = '17'</code> . Use straight quotes, not curly quotes.

Bar chart shows county codes not names	Bars labeled 17001, 17003 instead of county names	Remove FIPS from Rows. Drag County Name (the text column) to Rows instead.
Scatter plot shows one big dot	All geographic units collapse to a single point	In the Analysis menu, uncheck Aggregate Measures. Each geographic unit should now appear as a separate dot.
Filter actions not working	Clicking a geographic unit does not update other views	Dashboard, then Actions. Check that source and target sheet names match exactly. Sheet names are case-sensitive.
Trend line p-value differs from Script #7	Tableau shows different p-value than your R regression	Tableau uses all available rows. Script #7 used <code>drop_na()</code> . Note: The Tableau trend line includes all geographic units; the regression model used listwise deletion.
Point map for rate data	Each area has a dot at its center instead of shading, implying that the data applies to a single point.	Right-click FIPS → Geographic Role → County. In Marks card, confirm mark type is Map.
Choropleth map for facility locations	The entire geographic area is shaded to indicate that one clinic is present. This implies the entire area has that property.	For facility data (lat/lon CSV), drag Longitude to Columns, Latitude to Rows. Marks should select Circle, not Map.

8. Make Your Project Citable, Reproducible, and Findable

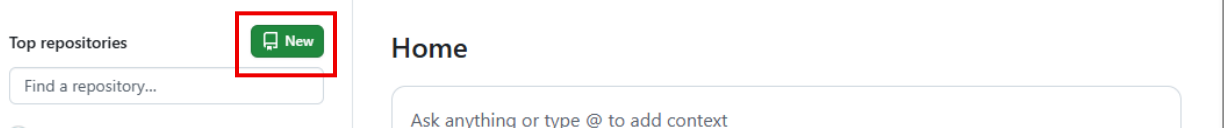
GitHub is a free platform for hosting, sharing, and version control of your research projects. Uploading your code, documentation, and shareable project files to GitHub supports reproducibility. Before beginning this section, you should have your completed project folder from Part 2, which should include R scripts, the master CSV, data dictionary, and README.txt.

Create a Free GitHub Account

1. Go to github.com and click Sign up in the top right corner.
2. On the registration page, click the **Continue with Google** button to link your Gmail account. Choose a professional GitHub username since this handle will be visible on your CV, resume, and LinkedIn profile (for example, hpotter-uis or harry-potter).
3. Complete the brief onboarding steps, ensuring you select the Free tier if prompted, and your new account is ready to go.

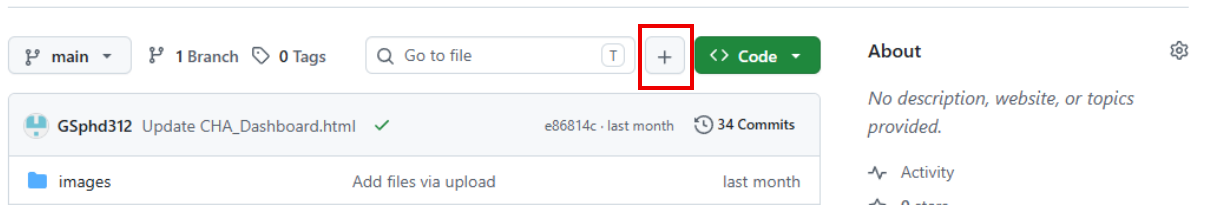
Create a New Repository

1. A repository (repo) is a folder on GitHub that holds all your project files. Click the **New** icon in the top right corner of GitHub to create a new repository.
2. Fill in the repository details with the repository name in lowercase (no spaces) and check the **Choose visibility*** set to Public. Then, click **Create repository**. Your empty repository is now live at: github.com/your-username/your-repository-name



Upload Your Project Files

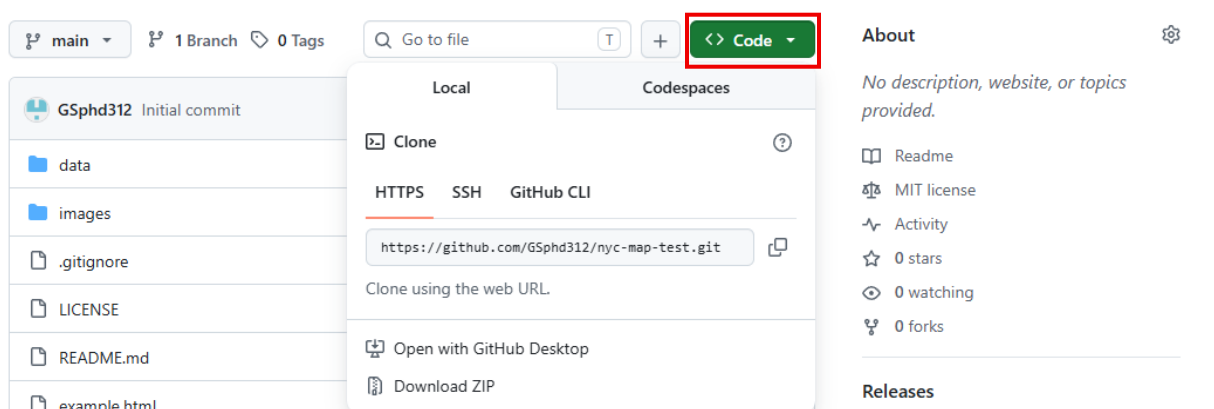
1. On your repository page, click **+**, then select Upload files.



- From your project folder, select and drag the following files: all R scripts from your code/ folder, your master CSV, your Tableau CSV, your data dictionary, and your README.txt. Then, click the **Commit changes** button. All your files are now live in your public repository.

Copy Your Repository URL

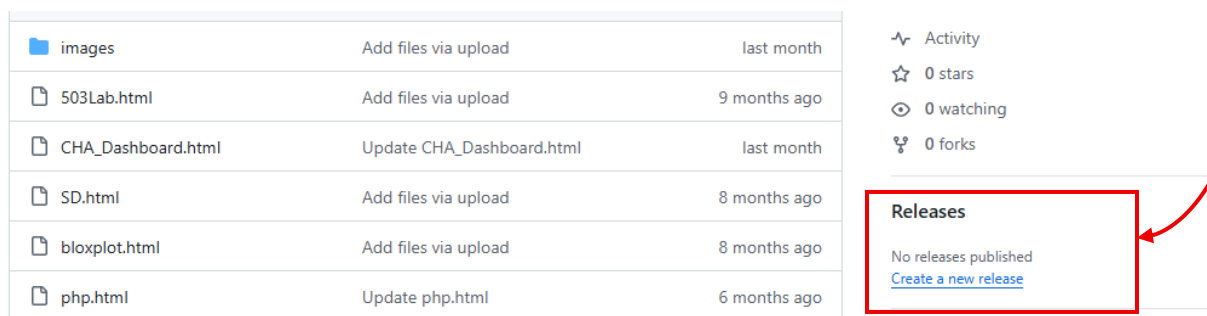
- Click the green <> **Code** button on your repository page and copy the URL shown. This is your GitHub repository URL.



- Paste it into your README.txt under the GitHub repository URL.

Register a DOI with Zenodo

- Go to zenodo.org and click **Log in with GitHub** to connect your accounts automatically.
- Once logged in, click your username in the top right corner of Zenodo and select **GitHub** from the dropdown menu. Hit the **Sync now** button to update the list from GitHub. Then, find your toolkit repository in the list and toggle it to **ON**.
- Back in GitHub, click **Releases** on the right side of the repository page, then click **Create a new release**. Click **Tag: Select tag**, then select **Create new tag**. Type v1.0, set the **Release title** to SDOH Open Scholarship Toolkit, [State]. Click **Publish release**.



New release

Tag: v1.0 Target: main

Excellent! This tag will be created from the target when you publish this release.

Release title
SDOH Open Scholarship Toolkit, [State].

Release notes
Select a previous tag to create generated release notes
Previous tag: Auto Generate release notes

Write Preview H B I $\equiv$ <>

Tagging suggestions
It's common practice to prefix your version names with the letter v. Some good tag names might be v1.0.0 or v2.3.4.

If the tag isn't meant for production use, add a pre-release version after the version name. Some good pre-release versions might be v0.2.0-alpha or v5.9-beta.3.

Semantic versioning
If you're new to releasing software, we highly recommend to [learn more about](#)

- Zenodo archives the release and creates a citable record. Processing time may vary. It will look like <https://doi.org/10.5281/zenodo.XXXXXXX>.
- Copy your DOI link and add it in two places: at the top of your README.txt file and in the description of your Tableau Public dashboard.



Quick Conceptual Video: [How to Make Your GitHub Code Citable with Zenodo | pyOpenSci](#). The button layouts on screen may vary slightly from the current live site.

Institutional Repository Sharing (Optional)

An institutional repository provides a university-supported location for preserving and sharing scholarly work. IDEALS (Illinois Digital Environment for Access to Learning and Scholarship) is the University of Illinois institutional repository and is one example. Learners affiliated with other institutions may use their own institutional repository or another appropriate open repository.

- For UIS-affiliated learners, submitting your project to IDEALS adds it to the UIS Collection of research, making it permanently accessible and linkable from your resume and LinkedIn.
- Unlike Zenodo, IDEALS does not assign a DOI but provides a permanent URL hosted by the university.
- To submit, complete the [UIS form](#). The UIS library will follow up when your submission is available or if they have questions. For more info, contact the UIS Library.

Side-by-Side Comparison of IDEALS and Zenodo

	IDEALS	Zenodo
What it is	UIS institutional repository	International Open Research Repository (CERN)
Who runs it	University of Illinois	CERN and OpenAIRE
DOI	No, permanent URL only	Yes, DOI is assigned automatically
What you deposit	Presentations/posters/reports	Datasets, code, any research output
How to submit	UIS form with librarian reviews and uploads.	Self-service
Best for	Presentations/posters/reports	Datasets, R scripts, GitHub repositories
UIS connection	Official UIS collection, institutional identity	No UIS branding
Voluntary	Yes	Yes

Your project is now fully open and FAIR:

FAIR	What You Did	Where in This Toolkit	Open Scholarship Requirement
------	--------------	-----------------------	------------------------------

Findable	GitHub README, Zenodo DOI, and IDEALS permanent URL	Part 3: GitHub, Zenodo IDEALS	Share your work openly
Accessible	Tableau Public dashboard, public GitHub repository, CSV format	Part 3: Tableau Public & GitHub	Share your work openly
Interoperable	FIPS codes for merging, standard column names, open file formats	Part 1 and Part 2	Make it reproducible
Reusable	Commented R scripts, codebook, data citations with download dates	Part 2	Cite your data and make it reproducible

9. Part 3 Checklists

- ☐ Check every item before submitting your Tableau Public URL. Complete all applicable items before submitting or sharing your Tableau Public URL is:
public.tableau.com/views/_____
- ☐ I have a Tableau Public account and am signed in at public.tableau.com.
- ☐ My master CSV (sdoh_[state]_tableau.csv) is uploaded and FIPS is set to String (Abc).
- ☐ I have built at least one Asset Map view (Simple Map or Point Distribution Map).
- ☐ I have built at least one Thematic Map view. The Filled (Choropleth) Map is required. Proportional Symbol, Dual-Axis, and Filled Map with Pie Charts are optional.
- ☐ I have assembled a Dashboard with at least three coordinated views.
- ☐ My choropleth map uses the correct color palette (sequential or diverging — not rainbow).
- ☐ Filter actions are working — clicking the map updates the bar chart and scatter plot.
- ☐ Every sheet has a descriptive title (not Sheet 1 or Map).
- ☐ The dashboard has a data source note at the bottom listing all datasets used.
- ☐ The tooltip on each view shows geographic unit name, outcome value, and at least one predictor value.
- ☐ I have built a Story Map with at least three story points linking my research question, findings, and conclusion.
- ☐ The dashboard is published to Tableau Public and set to Public visibility.
- ☐ Workbook downloads are enabled so others can adapt the template.
- ☐ The Tableau Public URL is recorded and ready to share.
- ☐ My GitHub repository contains R scripts, master CSV, data dictionary, and README.txt. If required for a course, I have submitted the repository URL according to my instructor's directions.
- ☐ When appropriate for my project, I have created a citable Zenodo record and added the DOI or persistent link to my README and Tableau Public dashboard description.
- ☐ If appropriate, I have deposited or shared my work through an institutional or open repository.

✓ **Your dashboard is published, publicly accessible, and documented for reuse. That is open scholarship.**

Part 1: SDOH Data Orientation · Part 2: Download & R Workflow · Part 3: Tableau, Repository & Open Sharing